

Electricity Efficiency Challenge - STEM

Storyline Project Grade Level: 5–6

Lesson Title	Driving Question	Lesson Summary	Lesson Outcomes	Learning Tasks	Assessment Tasks	Resources
What is Electricity?	What is electricity and where does it come from?	Students are introduced to the basic concepts of electricity, including its types (static/current) and sources (renewable/non-renewable).	Understand what electricity is. Identify various sources of electricity. Differentiate between renewable and non-renewable sources.	Brainstorm electricity usage at school and home. Watch a short video on electricity sources. Complete a worksheet comparing energy sources.	Electricity concept worksheet Group discussion reflection	Video clips Worksheets Images of power plants, solar panels, etc.
Why is Electricity Wasted?	What causes electricity to be wasted and why is it important to save it?	Students explore everyday behaviors and devices that lead to energy waste and analyze their impacts.	Identify common causes of electricity wastage. Explain the effects of wastage on the environment and economy.	Conduct a classroom energy audit. Create a list of energy-wasting habits. Graph electricity use patterns (math integration).	Group report on findings Energy audit checklist	Energy audit checklist Graph paper or tablets Real school/home electricity bills (optional)
How Do Circuits Work?	How does electricity flow in a circuit?	Students engage in a hands-on activity to build simple circuits, learning about current	Build a complete circuit. Identify conductors and insulators. Understand	Construct circuits using wires, batteries, bulbs. Experiment with	Circuit-building worksheet Peer demonstration	Bulbs, wires, batteries, switches Circuit diagrams Science journals

		flow, conductors, and insulators.	the function of switches and batteries.	different materials. Record observations.		
Design an Energy-Saving Device	What kind of device can help us reduce electricity waste?	Students brainstorm and plan devices (e.g., light timers, insulation ideas, reminders) that could help reduce energy use.	Collaborate to generate creative design ideas. Sketch and label a device plan. Justify design choices.	Brainstorm in teams Sketch and explain a design Peer feedback on feasibility	Design sketches with labels Team presentation of concept	Chart paper, markers Sample prototypes Engineering notebooks
Build and Test Your Device	How well does our device work and how can we improve it?	Students build their energy-saving device using simple materials, test it, and make necessary improvements.	Construct a working model. Test for functionality. Revise design based on test data.	Build device using recycled and simple materials Test performance (e.g., time light stays on) Document results	Test data sheet Improvement journal entry	Recycled materials (bottles, cardboard, tape) Multimeters (if available) Observational checklist
Present Your Solution	How can we share our energy-saving idea effectively?	Students present their final devices to peers and explain how they work to reduce electricity usage.	Demonstrate understanding of the problem and solution. Communicate design and impact clearly. Reflect on learning and improvements.	Prepare poster or digital slides Rehearse group presentation Peer Q&A session	Presentation rubric Self- and peer-assessment forms	Presentation templates (paper or digital) Projector (optional) Reflection sheets